Airport Network Design Using VLANs

Router-on-a-Stick Inter-VLAN Routing
with Cisco Packet Tracer

Topic	Airport Network Segmentation Using VLANs
Technique	Router-on-a-Stick Inter-VLAN Routing
Tool	Cisco Packet Tracer
VLANs	7 (Admin, Check-In, Security, Baggage, ATC, Server, WiFi)
Devices	1 Router, 2 Switches, 10 PCs, 3 Servers, 1 Access Point

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1. Introduction

Modern airports are complex environments that require a robust and well-segmented network infrastructure to support a wide range of operations simultaneously. From administrative tasks and passenger check-in to baggage handling, air traffic control, and public Wi-Fi services, each department demands reliable, isolated, and efficient network communication.

This project presents the design and implementation of a segmented airport network using Virtual Local Area Networks (VLANs) in Cisco Packet Tracer. The network employs the Router-on-a-Stick technique to facilitate inter-VLAN communication through a single physical router interface using logical subinterfaces. The design includes seven VLANs representing distinct airport departments, supported by centralized DHCP, DNS, and Web server infrastructure.

2. Objectives

1. Design a multi-VLAN airport network that logically segments departments for security and efficiency.
2. Implement Router-on-a-Stick (Inter-VLAN Routing) using a Cisco 2911 router with subinterfaces.
3. Configure VLAN trunk links between switches to carry traffic from multiple VLANs.
4. Set up a centralized DHCP server to automatically assign IP addresses to network devices.
5. Configure a DNS server to resolve domain names and a Web server to host an internal website.
6. Test and verify network connectivity both within the same VLAN and across different VLANs.
7. Simulate the complete network using Cisco Packet Tracer.

3. Background Study

Virtual LAN (VLAN):

A VLAN is a logical subdivision of a network that groups devices regardless of their physical location. VLANs improve network security by isolating traffic between departments, reduce broadcast domains, and simplify network management. Each VLAN behaves like an independent network segment even if the devices share the same physical switch.

Trunk Links:

A trunk link is a point-to-point link between two network devices that carries traffic for multiple VLANs. IEEE 802.1Q encapsulation is used to tag frames with their respective VLAN IDs as

they traverse the trunk. Trunk ports are configured between switches and between a switch and a router to allow inter-VLAN communication.

Router-on-a-Stick:

Router-on-a-Stick is a method of inter-VLAN routing that uses a single physical router interface divided into multiple logical subinterfaces. Each subinterface is assigned an IP address that serves as the default gateway for a specific VLAN. This allows devices on different VLANs to communicate through the router without requiring multiple physical interfaces.

DHCP, DNS, and Web Services:

The Dynamic Host Configuration Protocol (DHCP) automates IP address assignment. The Domain Name System (DNS) resolves human-readable hostnames to IP addresses. A web server hosts internal web pages accessible to users on the network.

4. Methodology

The project was implemented using Cisco Packet Tracer following a structured step-by-step approach:

Network Planning: Identified seven airport departments and assigned each a unique VLAN ID, name, and IP address range.

Topology Design: Designed the physical and logical topology using two 2960 switches, one 2911 router, servers, PCs, and an access point.

VLAN Configuration: Created VLANs on both switches and assigned access ports to the appropriate VLANs.

Trunk Configuration: Configured trunk links between Switch-1 and Switch-2, and between Switch-1 and the router.

Router Configuration: Configured subinterfaces on the router's GigabitEthernet0/0 interface for each VLAN using 802.1Q encapsulation.

Server Setup: Configured DHCP pools for each VLAN, DNS A-records, and the HTTP/HTTPS web server.

End Device Configuration: Assigned static IP addresses to servers and configured PCs either statically or via DHCP.

Testing: Verified connectivity using ping tests within the same VLAN and across different VLANs.

5. Network Design

5.1 VLAN Structure

VLAN ID	Name	Network Address	Gateway	Devices
10	ADMIN	192.168.10.0/24	192.168.10.1	PC0, PC1
20	CHECKIN	192.168.20.0/24	192.168.20.1	PC2, PC3
30	SECURITY	192.168.30.0/24	192.168.30.1	PC4, PC5
40	BAGGAGE	192.168.40.0/24	192.168.40.1	PC6, PC7
50	ATC	192.168.50.0/24	192.168.50.1	PC8, PC9
60	SERVER	192.168.60.0/24	192.168.60.1	DHCP, DNS, Web
70	WIFI	192.168.70.0/24	192.168.70.1	Access Point

5.2 Network Topology

The network consists of one Cisco 2911 Router, two Cisco 2960 Switches, ten PCs, three servers (DHCP, DNS, Web), and one wireless access point. Switch-1 connects to the router and to Switch-2 via trunk links. Switch-2 connects all department PCs.

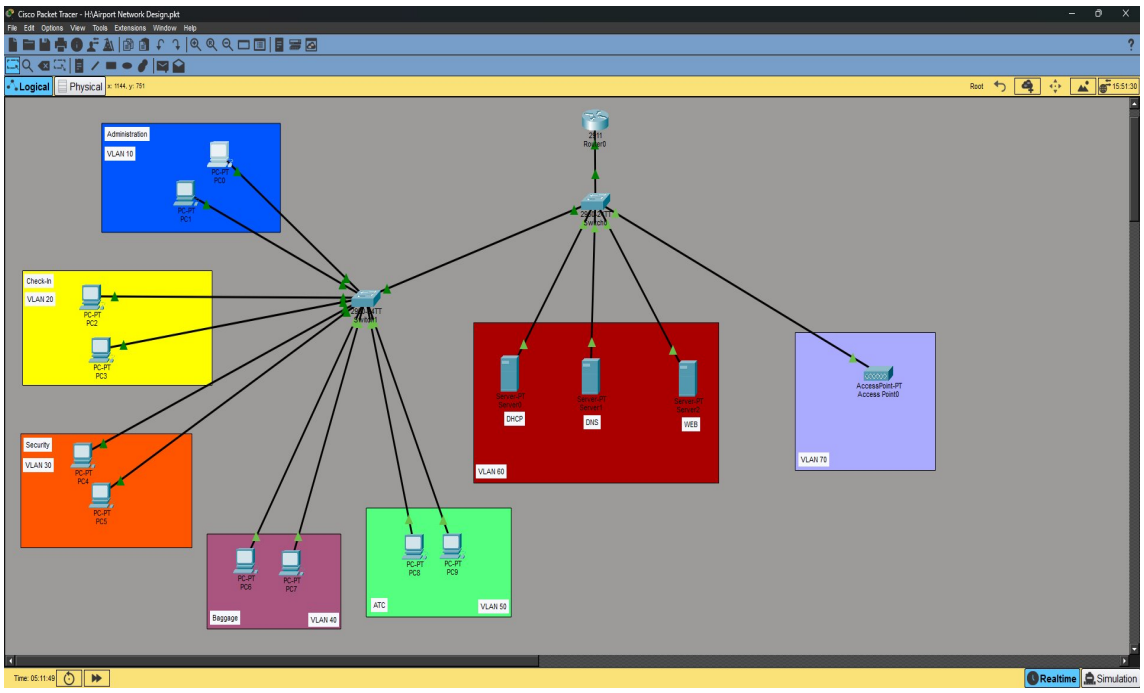


Figure 1: Airport Network Topology in Cisco Packet Tracer

6. Hardware and Software Requirements

Component	Model/Type	Quantity	Purpose
Router	Cisco 2911	1	Inter-VLAN routing (Router-on-a-Stick)
Switch (Distribution)	Cisco 2960-24TT	1	Connects router, servers, AP
Switch (Access)	Cisco 2960-24TT	1	Connects department PCs
PCs	PC-PT	10	End-user devices (2 per VLAN)
DHCP Server	Server-PT	1	Assigns IP addresses to VLANs
DNS Server	Server-PT	1	Resolves www.airport.com
Web Server	Server-PT	1	Hosts internal web pages
Access Point	AccessPoint-PT	1	Provides VLAN 70 wireless access
Software	Cisco Packet Tracer	1	Network simulation environment

7. Implementation

7.1 VLAN Creation

VLANs were created on both switches using the global configuration mode. Each VLAN was assigned a unique ID and a descriptive name corresponding to its airport department. The screenshot below shows the VLAN configuration on Switch-2.

```
Switch>show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                 Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                 Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                 Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                 Fa0/22, Fa0/23, Gig0/1, Gig0/2
10   ADMIN                             active    Fa0/1
20   CHECKIN                           active    Fa0/2
30   SECURITY                          active    Fa0/3
40   BAGGAGE                           active    Fa0/4
50   ATC                               active    Fa0/5
60   SERVER                            active
70   WIFI                              active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
```

Figure 2: VLAN Creation on Switch-2 (show vlan brief)

7.2 Port Assignment

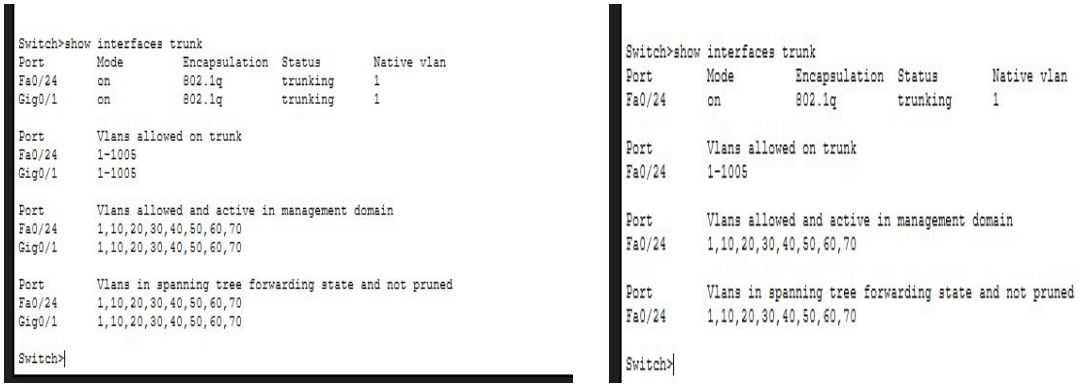
Access ports on Switch-2 were assigned to their respective VLANs. Each pair of PCs was connected to adjacent ports configured as access ports for the corresponding VLAN. Ports Fa0/1–Fa0/10 were assigned to VLANs 10 through 50.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/6
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/7
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/8
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/9
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/10
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#end
```

Figure 3: Port Assignment Configuration on Switch-2

7.3 Trunk Configuration

Trunk links were configured between Switch-1 and Switch-2 (Fa0/24) and between Switch-1 and the router (GigabitEthernet0/1). Trunk ports carry tagged traffic for all VLANs using 802.1Q encapsulation.



```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,10,20,30,40,50,60,70
Gig0/1      1,10,20,30,40,50,60,70

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,10,20,30,40,50,60,70
Gig0/1      1,10,20,30,40,50,60,70

Switch>
```

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,10,20,30,40,50,60,70

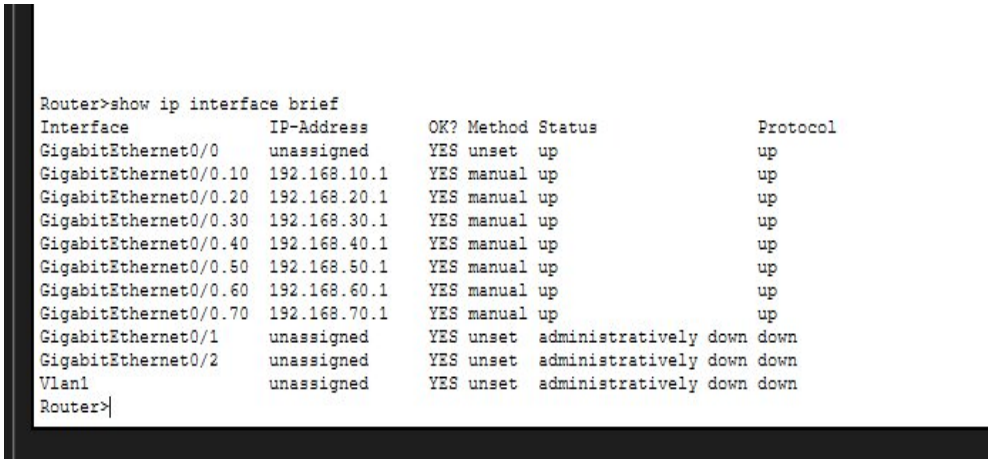
Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,10,20,30,40,50,60,70

Switch>
```

Figure 4: Trunk Configuration — Switch-1 (left) and Switch-2 (right)

7.4 Router-on-a-Stick Configuration

The Cisco 2911 router was configured with subinterfaces on GigabitEthernet0/0. Each subinterface was assigned an IP address as the default gateway for its VLAN, with 802.1Q encapsulation specifying the VLAN ID.



```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/0.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/0.60  192.168.60.1    YES manual up                    up
GigabitEthernet0/0.70  192.168.70.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

Figure 5: Router Subinterface Configuration (show ip interface brief)

7.5 DHCP Server Configuration

The DHCP server (192.168.60.10) was configured with separate address pools for each VLAN. Each pool specifies the default gateway, DNS server address, starting IP address, and subnet mask to automatically assign addresses to end devices.

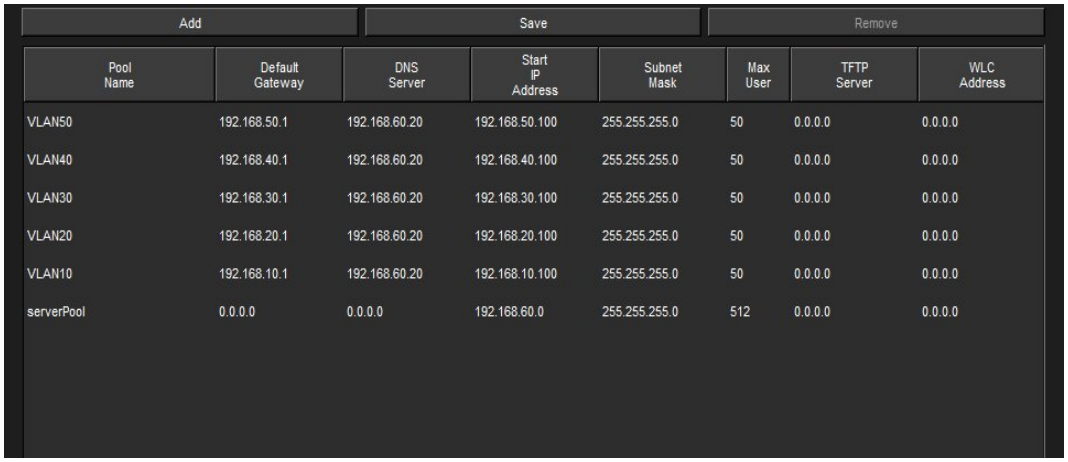


Figure 6: DHCP Server Pool Configuration

7.6 DNS Server Configuration

The DNS server (192.168.60.20) was configured with an A Record mapping www.airport.com to the Web Server's IP address (192.168.60.30), enabling name-based access to the internal web service.

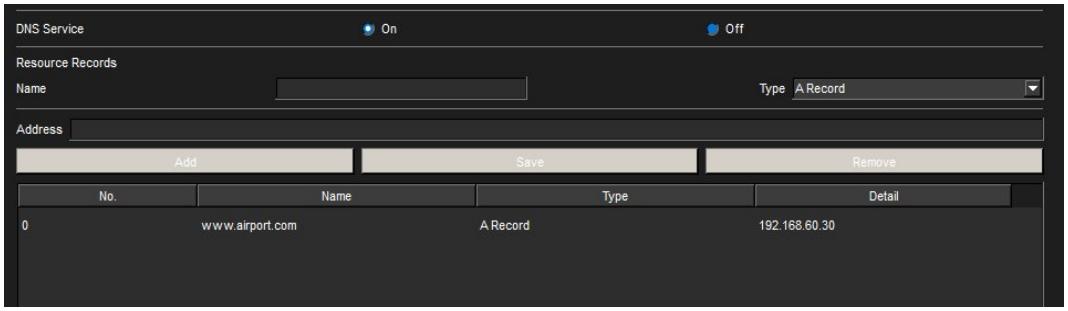


Figure 7: DNS Server Configuration

7.7 Web Server Configuration

The Web server (192.168.60.30) was configured with HTTP and HTTPS services enabled. The default index.html page was available for internal users to access via browser using the domain name www.airport.com.

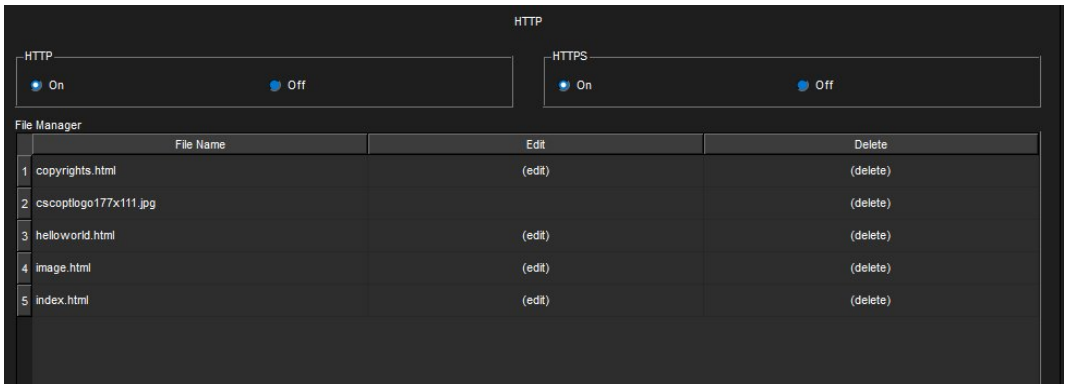


Figure 8: Web Server HTTP/HTTPS Configuration

8. Testing and Results

Connectivity was tested using ICMP ping commands from PC0 to verify both intra-VLAN and inter-VLAN communication. The following tests were performed:

Test	Source	Destination	Result
Same VLAN Ping	PC0 (192.168.10.10)	PC1 (192.168.10.11)	4/4 Success (0% loss)
Inter-VLAN Ping	PC0 (192.168.10.10)	PC2 (192.168.20.10)	3/4 Success*
Server Ping	PC0 (192.168.10.10)	Web Server (192.168.60.30)	4/4 Success (0% loss)
Gateway Ping	PC0 (192.168.10.10)	Gateway (192.168.10.1)	4/4 Success (0% loss)

* First packet timed out due to ARP resolution — this is normal behavior in Cisco Packet Tracer.

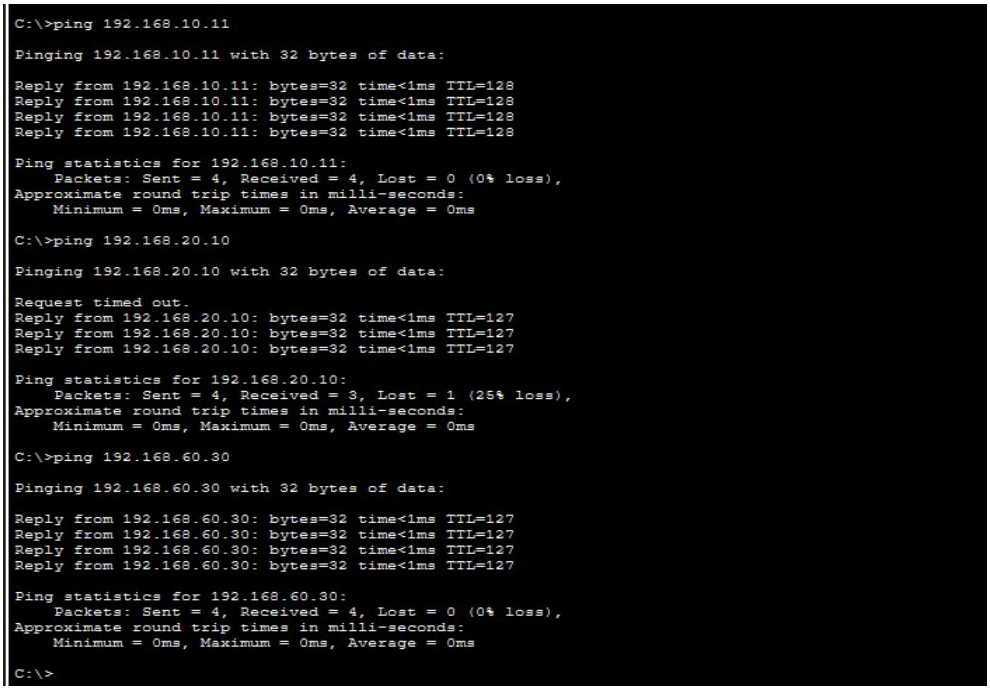


Figure 9: Ping Test Results from PC0

9. Challenges Faced

VLAN Propagation Issue: VLANs created on Switch-2 were not automatically propagated to Switch-1. This caused trunk ports to show only a subset of active VLANs. The issue was resolved by manually creating all required VLANs on Switch-1 as well.

Incorrect Trunk Port on Switch-1: Initially, FastEthernet0/1 was configured as a trunk port on Switch-1, but the router was actually connected to GigabitEthernet0/1. Reconfiguring Gi0/1 as a trunk resolved inter-VLAN routing failures.

PC Port Mismatch: Several PCs were connected to switch ports that belonged to different VLANs than intended. Systematic verification of each PC's cable connection and reassignment of ports corrected the connectivity issues.

Initial Ping Timeout: The first ICMP packet in each ping session timed out due to ARP cache being empty. This is standard behavior in Packet Tracer and not an actual network failure.

Session Timeout During Configuration: The Cisco Packet Tracer CLI session timed out during long configuration processes, requiring reconnection and re-entry of the enable and configure terminal commands.

10. Future Improvements

1. Implement VLAN Trunking Protocol (VTP) to automate VLAN propagation across multiple switches.
2. Replace Router-on-a-Stick with a Layer 3 switch for higher-performance inter-VLAN routing.
3. Add redundancy using Spanning Tree Protocol (STP) and redundant uplinks between switches.
4. Implement Access Control Lists (ACLs) to enforce security policies between VLANs.
5. Configure port security on switch access ports to prevent unauthorized device connections.
6. Deploy WPA2-Enterprise authentication for the wireless VLAN using a RADIUS server.
7. Expand the network to include additional airport areas such as cargo, lounges, and gates.
8. Integrate Quality of Service (QoS) policies to prioritize ATC and critical system traffic.

11. Conclusion

This project successfully demonstrated the design and implementation of a segmented airport network using VLANs and Router-on-a-Stick inter-VLAN routing in Cisco Packet Tracer. Seven VLANs were created to isolate traffic between airport departments including Administration, Check-In, Security, Baggage, ATC, Server infrastructure, and Wireless access.

The implementation of trunk links between switches and subinterfaces on the router enabled seamless communication between all VLANs while maintaining logical separation. Centralized DHCP, DNS, and Web server services were successfully deployed in VLAN 60 and made accessible to all network segments.

Connectivity tests confirmed that both intra-VLAN and inter-VLAN communication functioned correctly. Several real-world networking challenges were encountered and resolved during the implementation, providing valuable hands-on experience with enterprise network configuration and troubleshooting.

12. References

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